Graduation Thesis Outline & Review Feedback Guide

SynapseKT: Dual-Process Cognitive Modeling and 2PL Psychometrics for Attentive Knowledge Tracing

Candidate: Amirbek Abdirasulov | **Supervisor:** Prof. Yeon-Chang Lee
Date: October 31, 2026 | **Milestone:** End-of-October Full Thesis Draft
Project Repository & Reproducibility: https://github.com/mirabdi/KT

---

1 1. Topic Refinement & Academic Rationale

Initial Agreed Topic: *Knowledge-Graph-Guided Dynamic Learning Path Recommendation* (curriculum subgraph path prediction).

Rationale for Refinement to Deep Knowledge Tracing (KT):

- **Offline Evaluation & Benchmarking Bottleneck:** Dynamic path recommendation formulated as graph path prediction lacks standardized, reproducible offline evaluation benchmarks. Without continuous, long-term live deployments with human learner cohorts, evaluating whether a path recommendation is truly "optimal" relies on circular simulation heuristics and synthetic assumptions that cannot be objectively verified.
- **Prerequisite Cognitive Engine:** In personalized education, Knowledge Tracing is the core foundational predictive engine that underpins any dynamic path recommendation: an intelligent system cannot reliably adapt a learning trajectory without first accurately tracing student mastery and cognitive decay across concepts.
- **Empirical Rigor & Standardized Benchmarks:** Knowledge Tracing features rigorous, globally recognized public benchmark datasets (e.g., ASSISTments, Algebra2005, NIPS34, Statics2011) with verifiable chronological response logs, enabling strict 5-fold student cross-validation and reproducible scientific evaluation.

2 2. Thesis Structure & Technical Summary of SynapseKT

The full thesis draft is compiled in `papers/synapsekt/main.pdf` (ACL preprint format, 10 pages). It consists of seven structured sections:

2.1 Sections 1–2: Problem Formulation & The Sequence Modeling Dilemma

Online KT estimates sequential performance $\hat{r}_t = P(r_t = 1 \mid q_t, c_t, \mathcal{H}_{t-1})$ under strict causal constraints. We identify two core theoretical flaws across existing architectures:

1. **Momentum vs. Retrieval Dilemma:** Recurrent models (DKT, CKT) capture short-term streaks but compress history into a single vector, causing catastrophic forgetting on long histories. Transformers (SAKT, AKT, simpleKT) perform content-addressable historical retrieval but suffer from *attention dilution* on dense histories ($T > 500$), losing sensitivity to immediate momentum and fatigue.
2. **The Psychometric Gap:** Free dense question embeddings overfit long-tail items, while 1PL Rasch models assign only a single scalar difficulty (μ_q), ignoring that questions differ fundamentally in diagnostic discrimination power (α_q).

2.2 Section 3: The SynapseKT Cognitive Architecture

Drawing on dual-process cognitive psychology (Baddeley working memory + Ebbinghaus long-term memory), SynapseKT introduces:

- **2PL-Rasch Item Projections:** Jointly parameterizes item difficulty $\mu_q \in \mathbb{R}$ and bounded item discrimination $\alpha_q = 2\sigma(\nu_q) \in (0, 2)$ with empirical Bayes L_2 shrinkage ($\mathcal{R}_{\text{2PL}} = \lambda_\mu \sum \mu_q^2 + \lambda_\alpha \sum \nu_q^2$), preventing tail overfitting while sharpening diagnostic signal.
- **Stream A (Multi-Scale Working Memory):** Receptive field bank of causal 1D depthwise convolutions ($K \in \{3, 5, 7\}$) with Gated Linear Units (GLUs) over right-shifted past interactions, capturing session momentum and local streaks without state drift.
- **Stream B (Adaptive Monotonic Attention):** Queries long-term curriculum mastery with an **Adaptive Ebbinghaus Consolidation Kernel**. The forgetting decay rate $\theta_{h,t} = \text{Softplus}(\gamma_h)/(1 + \log(1 + N_{c_t,<t}))$ dynamically flattens as concept practice count $N_{c_t,<t}$ increases, modeling the cognitive *spacing effect*.
- **Learned Cognitive Confidence Gate:** A feature-wise soft multiplexer $\mathbf{g}_t \in (0, 1)^D$ that dynamically balances working memory momentum against long-term concept recall.

2.3 Sections 4–5: Benchmark Evaluation & Empirical Breakthroughs

Evaluated across **6 standard benchmarks** against **19 competitive baselines** under strict 5-fold student cross-validation on our standardized `ktbench` framework (Table 1):

- **Longitudinal Tutoring Breakthrough:** SynapseKT achieves **0.8200 AUC** on ASSISTments2017—a **+5.28% leap** over the strongest prior attentive model (AKT-R, 0.7672), overcoming attention smearing on long trajectories ($L = 551$).
- **State-of-the-Art Dominance:** Establishes top AUC across ASSISTments2009 (**0.8384**), ASSISTments2015 (**0.7857**), and NIPS34 (**0.7896**), while scaling gracefully to extreme catalogs (Algebra2005: 173k items, **0.9187 AUC**).

Model Architecture	ASSIST2009	ASSIST2015	ASSIST2017	Algebra2005	NIPS34	Statics2011
SynapseKT (Ours)	**0.8384 ± 0.004**	**0.7857 ± 0.002**	**0.8200 ± 0.004**	0.9187 ± 0.007	**0.7896 ± 0.001**	0.8270 ± 0.004
AKT-R	0.8361 ± 0.004	—	0.7672 ± 0.004	0.9167 ± 0.009	0.6661 ± 0.081	—
DKT-R / DKT+	0.8189 ± 0.004	0.7322 ± 0.002	0.7606 ± 0.004	0.9183 ± 0.000	0.7889 ± 0.000	**0.8294 ± 0.004**
DKVMN-R	0.8247 ± 0.004	0.7273 ± 0.002	0.7572 ± 0.003	0.9185 ± 0.000	0.7891 ± 0.000	0.8215 ± 0.004
SIMPLEKT	0.7549	0.6997	0.7087	0.8838	0.7826	0.6446

Table 1: Test AUC comparison across standard educational benchmarks (5-fold student cross-validation).

2.4 Sections 6–7: Ablation Analysis, Hardware Profiling & Deployment

- **Ablation Matrix (15 Variants):** Removing working memory (`-NoWM`) causes a steep -1.46% **AUC drop** (0.7969 to 0.7823), verifying that local momentum and long-term retrieval provide complementary signals.
- **Edge & Hardware Efficiency:** Peak unified memory is bounded at **2.04 GB** on Apple Silicon with **3.82 ms** single-step latency, satisfying real-time ITS SLAs (< 10 ms).

3 3. Summary of Deliverables Submitted

1. **Full Graduation Thesis Draft:** `papers/synapsekt/main.pdf` (Complete research manuscript, ACL style).
2. **Comprehensive Survey Report (1-Credit Internship):** `papers/kt_survey/main.pdf` (13-page extensive survey and empirical benchmark across 6 KT paradigms, 20 architectures, and 390 runs).
3. **Open-Source Codebase:** Hosted at https://github.com/mirabdi/KT with modular PyTorch models, data loaders, evaluation pipelines, and test suites.

4 4. Targeted Points for Supervisor Review & Feedback

To optimize your review time, I would welcome your thoughts on four specific aspects:

1. **Topic Refinement Approval:** Does grounding the thesis in foundational Knowledge Tracing—enabling reproducible empirical evaluation across 6 benchmarks—satisfy your expectations for the graduation thesis and internship scope?
2. **2PL Psychometric Formalism:** In Section 3.1, 2PL discrimination is bounded as $\alpha_q = 2\sigma(\nu_q)$ with empirical Bayes shrinkage. Do you recommend incorporating additional psychometric comparisons (e.g., multidimensional IRT)?
3. **Bridge to Downstream Recommendation:** Section 7 describes how SynapseKT's calibrated mastery logits serve as the input state for adaptive curriculum path selection. Would you recommend expanding this into a concrete downstream recommendation case study in the thesis, or keeping the focus strictly on sequence modeling?
4. **Final Draft Priorities (Late Nov – Early Dec):** What specific theoretical or empirical aspects should be prioritized as I refine the manuscript toward the final submission?